Name of the Experiment: Verify the operation of basic logic gates (AND, OR, NOT)

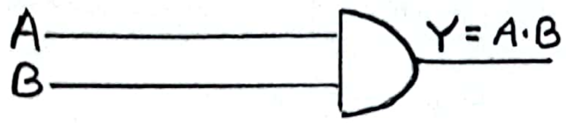
Objective: The objective of this expercimment is to verify the basic gates (AND, OR, NOT)

Theory: **Boolean Algebra:** Boolean Algebra differs in a major way from ordinary Algebra. Boolean Algebra is allowed to have any possible two values 0 or 1. In Boolean Algebra there are no function, negative, square, root, cube root and so on. In fact in Boolean Algebra, there are only three basic operators are called basic gates (AND, OR, NOT)

Logic gates : Digital circuit called basic gates which can be connected form diodes. transistor and resister are connected in such away that the circuit output is the result of a basic operation (AND, OR, NOT) performed on the output.

Basic logic gates :- Basic logic gates are those which have no other combination. There are three types of basic logic gates. The are :

i) AND - gates : Basic logic gates are An AND - gates which is a circuit that has two or more inputs and whose output is equal to the AND operation of the input. If the input A and B are logic operation of the output Y is a voltage which value is the result of the AND operation on A and B that is $Y = A \cdot B$



Truth Table:

Input		Output
A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

OR-Gate:

In digital circuit, OR-gate is a circuit that has two or more inputs and those output is equal to the OR operation of the input. If the inputs A and B are the logic gate operator and Y is a output then $Y = A + B$

CamScanner

Symbole:

Truth table :

Input		Output
0	0	0
0	1	1
1	0	1
1	1	1

NOT- Gate :

NOT - Gate is a digital circuit that has only one input and one output. It is also called INVERTER. It's output level is always opposite to the logic level of it's inputs. If the input A is logic voltage level whose value is the result of the NOT operation on A that is $Y = \overline{A}$

CS CamScanner

Symbole :

Truth Table :

Input	Output
A	$Y = \bar{A}$
0	1
1	0

Apparatus :

i) AND gate
ii) OR gate
iii) NOT gate
iv) Bread Board
v) Resistor
vi) Led
vii) Wire
viii) Voltage Supplier.

CS CamScanner

Procedure :

At first, we have to identify Pin number of IC. The IC is set on the bread board for AND gate. Pin 1 and 2 are input and pin 3 is output. So the wires are attached to the input pin. Another wire from output pin connected to the resister and the output Light of the bread board. We also connect the Vcc and GND with IC.

All IC's 7 and 14 pin is stand for GND and Vcc. Vcc is connected with 0V. And GND is connected with 0V. Then the power is applied to the bread board.

CS CamScanner

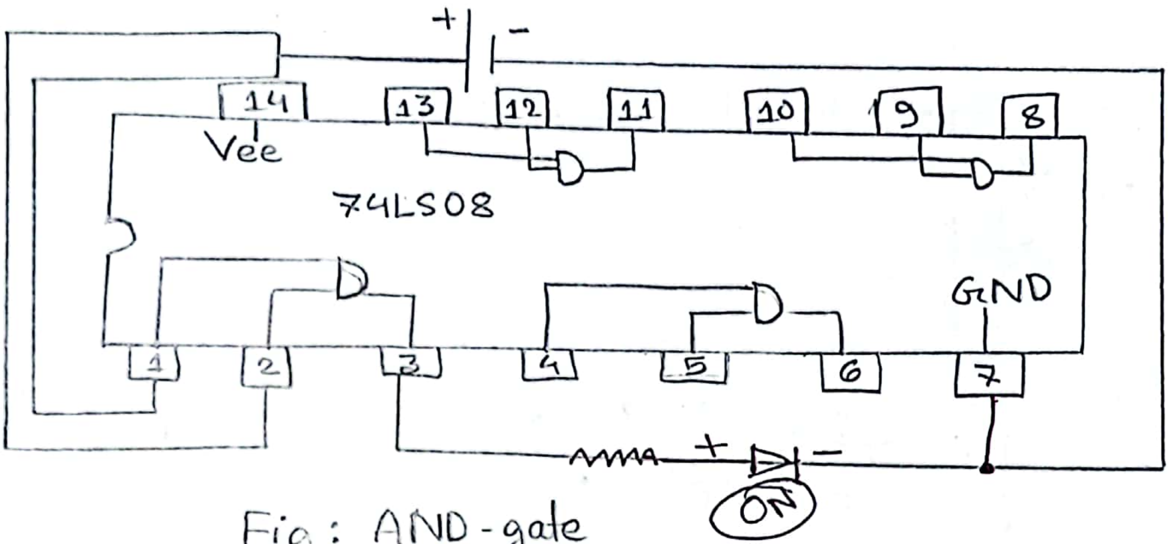


Fig: AND-gate

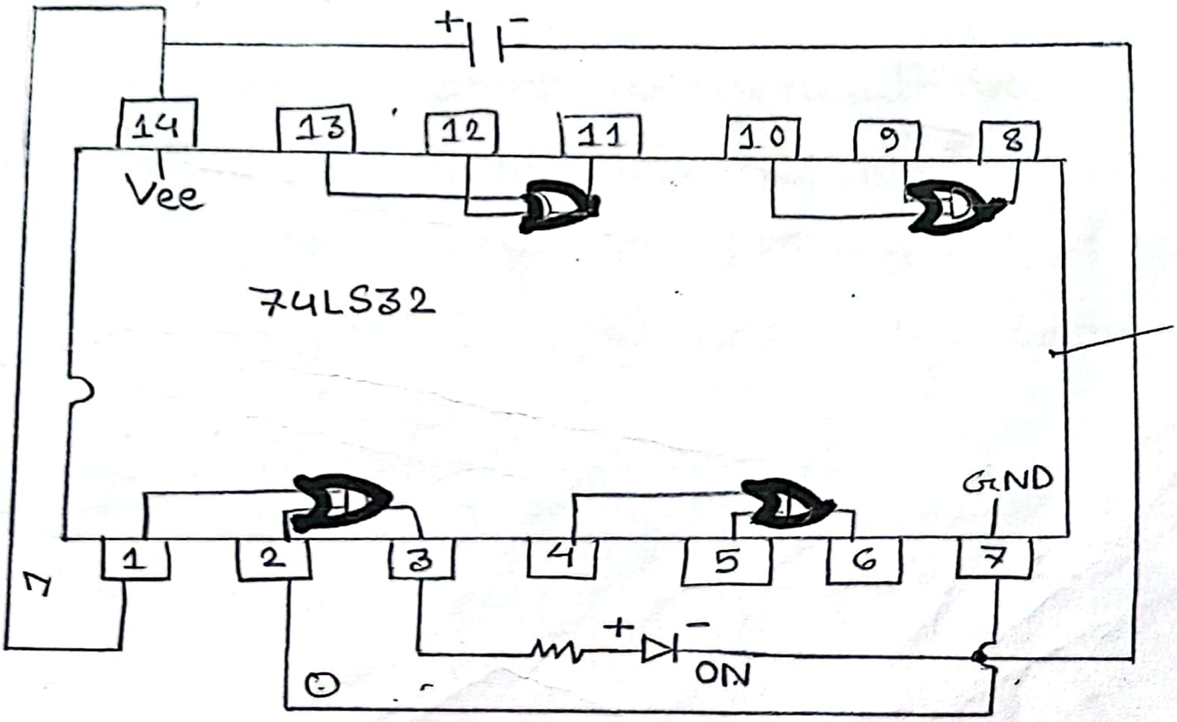


Fig: OR gate

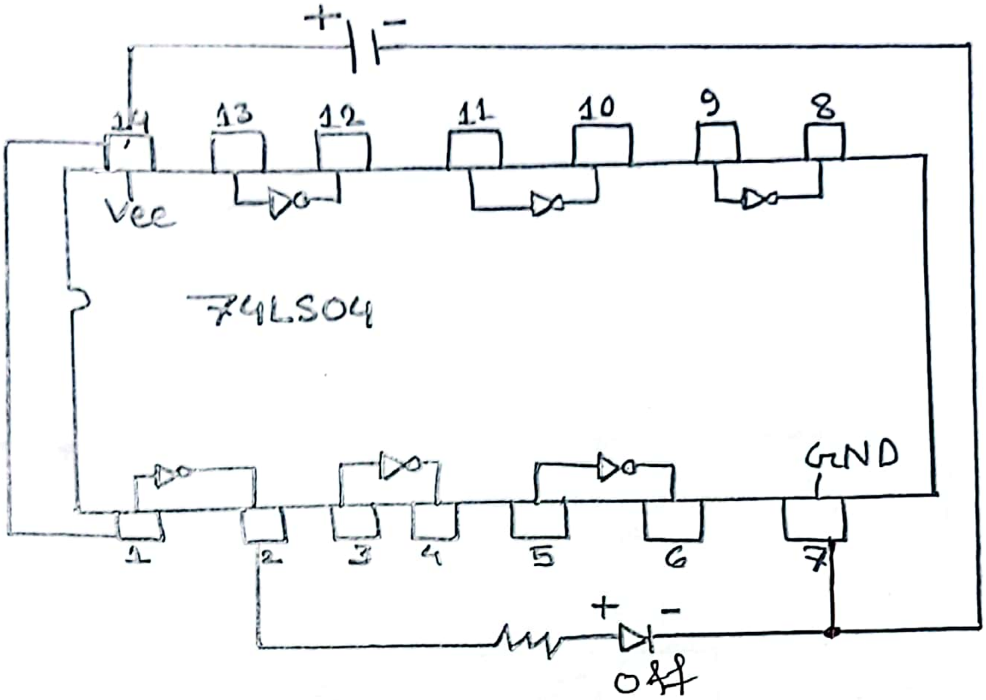


Fig: Not gate

When the input switch are on result are noted down. Thus we constructed the circuit for OR and NOT gate and get the possible output result for various possible input and noted down.

Result :

AND GATE

A	B	$Y = A \cdot B$	LED
0	0	0	off
0	1	0	off
1	0	0	off
1	1	1	on

CamScanner

OR-gate

A	B	Y = A+B	LED
0	0	0	off
0	1	1	on
1	0	1	on
1	1	1	on

NOT-gate

A	$Y = \overline{A}$	LED
0	1	on
1	0	off

Discussion :

We change all the gates of IC's and found the active gates. We found out the positions where the IC got proper supply. For this experiment, we use these IC's.

———— x ————